

Task 14: Interpretation of Evaluation Metrics

1. Mean Squared Error (MSE)

MSE measures the average of the squared differences between the actual and predicted values. Because errors are squared, larger deviations are penalized more heavily. A lower MSE indicates that the model predictions are closer to the true values, reflecting higher accuracy.

2. Mean Absolute Error (MAE)

MAE calculates the average of the absolute differences between actual and predicted values. Unlike MSE, it does not exaggerate large errors, making it easier to interpret. Since it is expressed in the same units as the target variable (INR), MAE directly shows the average prediction error magnitude. A lower MAE means the model consistently produces predictions close to actual house prices.

3. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE, bringing the error measure back to the same units as the target variable. It represents the typical size of prediction errors. A lower RMSE indicates that the model is more precise in predicting house prices. RMSE is often preferred because it balances interpretability with sensitivity to large errors.

4. R^2 Score (Coefficient of Determination)

R^2 explains the proportion of variance in the dependent variable (house price) that is accounted for by the independent variable(s). A value closer to 1 indicates a stronger model fit. For example, an R^2 of 0.82 means that 82% of the variation in house prices is explained by the predictor (area_sqft).

5. Adjusted R^2 Score

Adjusted R^2 modifies the R^2 value to account for the number of predictors in the model. In simple regression (with one predictor), R^2 and Adjusted R^2 are nearly identical. In multiple regression, Adjusted R^2 becomes crucial because it penalizes the inclusion of irrelevant predictors, ensuring that only meaningful variables improve the model's explanatory power.

Task 17: Explanation of Performance Change

◇ Why Performance Improves

- **More Predictors:** Multiple Linear Regression uses several features (area, bedrooms, bathrooms, location, age, lot size). This allows the model to capture more real-world factors influencing house price.
- **Higher R^2 / Adjusted R^2 :** With meaningful predictors, the model explains more variance in house prices, leading to better fit.
- **Lower Errors (MSE, MAE, RMSE):** Predictions become closer to actual values because the model considers multiple dimensions of housing characteristics.

◇ Why Performance May Degrade

- **Irrelevant Predictors:** If added features do not truly affect house price, they introduce noise.
- **Multicollinearity:** Strong correlation between predictors (e.g., bedrooms and bathrooms) can distort coefficient estimates.
- **Overfitting:** Adding too many variables may fit the training data too closely, reducing generalization to new data.
- **Adjusted R^2 Drop:** If predictors are weak, Adjusted R^2 decreases, signaling that the model complexity is not justified.

◇ Task 20: Identify Overfitting or Underfitting

- **Overfitting:**
 - Polynomial curve fits training data too closely (wiggly curve).
 - R^2 is very high on training data but poor on test data.
 - Happens with higher degrees (e.g., degree 5 or more).
- **Underfitting:**
 - Model fails to capture the relationship (straight line when data is curved).
 - Both training and test errors are high.

- Happens with very simple models (Linear Regression when data is nonlinear).

◇ Task 21: Conceptual Explanation of Gradient Descent

Gradient Descent is an optimization algorithm used to minimize a cost function (like Mean Squared Error in regression).

- It works by iteratively adjusting parameters (weights) in the opposite direction of the gradient of the cost function.
- **Learning Rate (α):** Controls the step size. Too high $\rightarrow$ overshoot, too low $\rightarrow$ slow convergence.
- Goal: Find the parameter values that minimize the cost function.

◇ Task 25: Compare Convergence & Training Time

- **Batch Gradient Descent:**
 - Uses all data per update.
 - Converges smoothly but can be slow on large datasets.
- **Stochastic Gradient Descent (SGD):**
 - Updates parameters for each sample.
 - Faster updates, but convergence path is noisy.
 - Good for very large datasets.
- **Mini-Batch Gradient Descent:**
 - Compromise between Batch and SGD.
 - Uses small batches (e.g., 32 samples).
 - Faster than Batch, smoother than SGD.
 - Most commonly used in practice (especially in deep learning).

